

ISYE 7406 HW3

Introduction

The primary objective of this study is to classify cars into two categories: high fuel efficiency ($\text{mpg01} = 1$) and low fuel efficiency ($\text{mpg01} = 0$), based on whether their MPG value is above or below the median. This classification allows us to identify the key vehicle features associated with fuel efficiency and develop predictive models for classifying future car models. To achieve this goal, classification techniques are employed, including Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), Naïve Bayes, Logistic Regression, and Principal Component Analysis K-Nearest Neighbors (PCA-KNN).

Exploratory Data Analysis

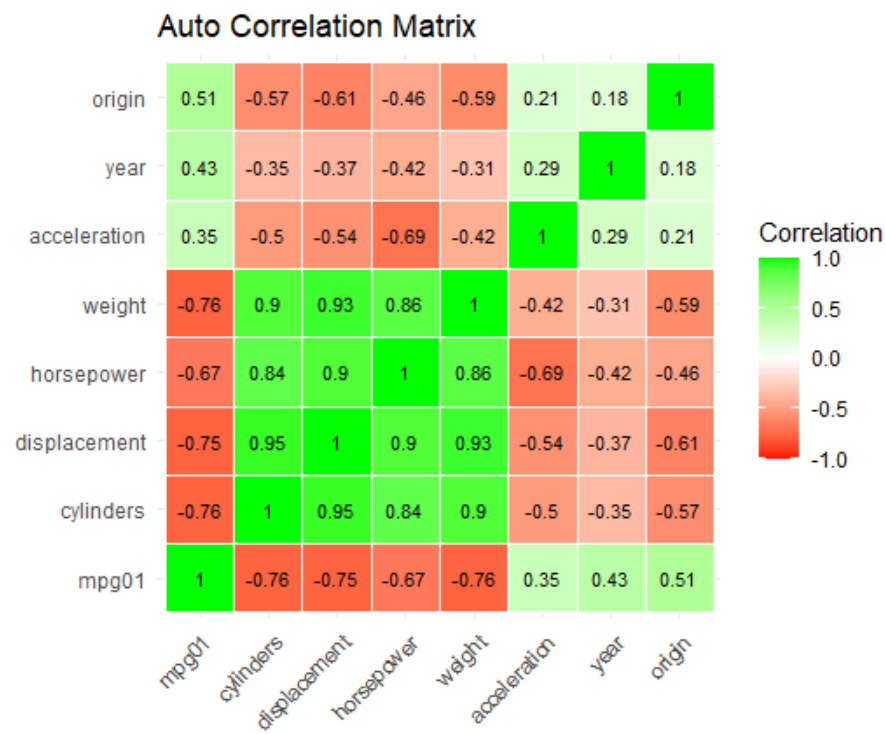
The Auto MPG dataset consists of vehicle characteristics that influence fuel efficiency. Key features include displacement, horsepower, weight, acceleration, cylinders, and origin, along with the target variable mpg (miles per gallon). To facilitate classification, we create a binary variable mpg01, where cars with MPG above the median are classified as high efficiency (1) and those below as low efficiency (0).

Figure 1 and 2: Scatterplots of displacement vs. horsepower, and weight vs. acceleration, and their classifications into mpg01



The summary statistics for the Auto MPG dataset provide perspective into the features that impact fuel efficiency. The mean MPG is 23.51, with a median of 23.0, indicating a relatively balanced distribution of fuel efficiency across vehicles. The standard deviation of 7.82 suggests moderate variation in MPG values.

Figure 3: Correlation Matrix of the Auto dataset



This correlation matrix illustrates the variables that are most associated with mpg01. The variables with a strong relationship—regardless of positive or inverse—are: cylinders, displacement, horsepower, and weight. This suggests that there is generally a profound effect on mpg01 if there are fewer cylinders, less total volume of cylinders (displacement), less engine power, and lighter weight. Conversely, acceleration, year and origin have a weaker correlation with mpg01, which is to be expected.

Table 1: Summary statistics of selected variables

Variable	Mean	Median	Standard Deviation
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MPG	23.51	23	7.82
Displacement	193.43	148.5	104.27
Horsepower	104.47	93.5	38.49
Weight	2970.42	2803.5	846.84
Acceleration	15.57	15.5	2.76

For engine size, displacement has an average of 193.43 with a wide spread (SD = 104.27), indicating diverse engine capacities in the dataset. Horsepower averages 104.47 but shows substantial variability (SD = 38.49), reflecting differences in vehicle power. Weight has a mean of 2970.42 pounds, with a significant spread (SD = 846.84), indicating cars with both light and heavy builds. As an example of a variable with a weaker relationship with mpg01, Acceleration averages 15.57 seconds, showing relatively small variation (SD = 2.76).

Methods

To classify vehicles based on their fuel efficiency, multiple machine learning techniques were employed, including Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), Naïve Bayes, Logistic Regression, and Principal Component Analysis K-Nearest Neighbors (PCA-KNN). Each method was tested on the dataset using relevant predictor variables: cylinders, displacement, horsepower, and weight.

- Linear Discriminant Analysis (LDA) – Projects the data into a space that maximizes class separation under Gaussian assumptions with a common covariance. It looks for a straight line (or plane in higher dimensions) that best separates the two classes, assuming both classes share the same overall spread.
- Quadratic Discriminant Analysis (QDA) – Similar to LDA but allows each class to have its own spread.
- Naive Bayes – Uses Bayes' Theorem and assumes that each feature is independent from each other.
- Logistic Regression – Models the log-odds of the binary outcome as a linear combination of predictors.

- PCA-KNN – Performs Principal Component Analysis (PCA) on the training features to reduce dimensionality, and then uses the K-Nearest Neighbors (KNN) algorithm in the reduced feature space. We also search for the optimal number of neighbors by comparing test errors for various k values.

After splitting the dataset into training and testing sets, we fit each model and record both training and testing error rates. To obtain more robust estimates of test performance, we also carry out Monte Carlo Cross-Validation (MCCV), repeating the random train–test splits 100 times and averaging the resulting test errors.

Results

Table 2: All models and their training and testing error (1 iteration)

Model	Training error	Testing error
LDA	0.1014493	0.06896552
QDA	0.08695652	0.06896552
Naive Bayes	0.09782609	0.09482759
Logistic Regression	0.09782609	0.05172414
PCA-KNN (3)	0.07246377	0.112069

Among the 5 models, logistic regression had the lowest testing error at 0.05172, which implies the best performing model, although its training error was the second highest and just behind LDA. PCA-KNN performed the worst among all models, but it did have the lowest training error.

Table 3: Finding optimal k-value

K	Testing error
1	0.1206897
3	0.1120690

5	0.1120690
7	0.1120690
9	0.1206897
11	0.1293103
13	0.1293103
15	0.1293103

A range of k values were observed to find the optimal k value by comparing their testing errors against one another. K = 3 is selected because it is the smallest k value with the lowest testing error.

Table 4: Monte Carlo Cross-Validation (100 iterations)

Model	Testing error
LDA	0.09508621
QDA	0.09655172
Naive Bayes	0.09905172
Logistic Regression	0.1016379
PCA-KNN (3)	0.1114655

A Monte Carlo Cross-Validation analysis is conducted. This is usually applicable to datasets with a weak ratio of observations to variables (when there are very few data points). The results showcase a difference in what is observed to be the best performing model, as LDA has the best performing model with a testing error of 0.09509

Findings

This analysis evaluated classification models: LDA, QDA, Naive Bayes, Logistic Regression, and PCA-KNN. These 5 models were utilized to predict whether a vehicle's mpg is above or below the median (1 or 0). Weakly-correlated factors were excluded from the models, and the models were assessed using single iteration and MCCV with 100 iterations to compare the strengths and limitations of each model.

For the single train-test split performance, Logistic Regression had the lowest testing error, and PCA-KNN had the highest testing error. With Monte Carlo Cross-Validation, LDA performed the best with the lowest testing error, and PCA-KNN remained as the worst performing model. This suggests that LDA may be more reliable across different train-test splits.